4. The unit of magnetic flux density is

   a) Weber

   b) weber/metre

   c) weber/meter$^2$

   d) weber . meter$^2$

## II. Fill in the blanks.

1. The SI Unit of magnetic field induction is _____.
2. Devices which is used to convert high alternating current to low alternating current is _____.
3. An electric motor converts _____.
4. A device for producing electric current is _____.

## III. Match the following.

| | |
|---|---|
| 1. Magnetic material | (a) Oersted |
| 2. Non-magnetic material | (b) Iron |
| 3. Current and magnetism | (c) Induction |
| 4. Electromagnetic induction | (d) Wood |
| 5. Electric generator | (e) Faraday |

## IV. State whether true or false. If false, correct the statement.

1. A generator converts mechanical energy into electrical energy.
2. Magnetic field lines always repel each other and do not intersect.
3. Fleming's Left hand rule is also known as Dynamo rule.
4. The speed of rotation of an electric motor can be increased by decreasing the area of the coil.
5. A transformer can step up direct current.
6. In a step down transformer the number of turns in primary coil is greater than that of the number of turns in the secondary coil.

## V. Answer in brief.

1. State Fleming's Left Hand Rule.
2. Define magnetic flux density.
3. List the main parts of an electric motor.
4. Draw and label the diagram of an AC generator.
5. State the advantages of ac over dc.
6. Differentiate step up and step down transformer.
7. A portable radio has a built in transformer so that it can work from the mains instead of batteries. Is this a step up or step down transformer? Give reason.
8. State Faraday's laws of electromagnetic induction.

## VI. Answer in detail.

1. Explain the principle, construction and working of a dc motor.
2. Explain two types of transformer.
3. Draw a neat diagram of an AC generator and explain its working.

## REFERENCE BOOKS


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Fundamental University Physics – M. Alonso, E. J. Finn Addimon Wesley (1967)


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